

the application.

Messages sent to a Node can be lost for various reasons such as lossy network or insufficient buffer space at the receiver. In the case of [Intermittently Connected Devices](#), which are active infrequently to receive messages destined for them, a sender must be aware of the characteristics of the recipient to ensure it does not attempt to send at a rate beyond the recipient's capability. Therefore, the sender SHALL choose retransmission timeouts based on the [session characteristics](#) of the destination Node exposed via [Section 4.3.2, "Operational Discovery"](#).

At each sender, a retransmission timer is started each time a reliable message is transmitted. The duration of the retransmission timer SHALL be calculated as follows:

```
"mrpBackoffTime" = i * "MRP_BACKOFF_BASE"^(max(0,n-"MRP_BACKOFF_THRESHOLD")) * (1.0 +
"random"(0,1) * "MRP_BACKOFF_JITTER")
```

Where:

```
{:("mrpBackoffTime", =, "the resultant retransmission timeout for this
transmission"),(n, =, "the number of send attempts before the current one for this
message (0 if this is the initial transmission)"),(i, =, "the base retry interval for
the Exchange (either IDLE or ACTIVE)":)}
```

For each unique Exchange, the sender SHALL wait for the acknowledgement message until the retransmission timer, *mrpBackoffTime*, expires.

When waiting for an acknowledgement, an [Intermittently Connected Device](#) will always be in [Active Mode](#) since it will have at least one active exchange. An [Intermittently Connected Device](#) sender SHOULD increase the *mrpBackoffTime* by its [fast polling interval](#) to take into account the delay that might happen in receiving the acknowledgment while in [Active Mode](#).

For the first message of a new exchange, the base interval, *i*, SHALL be set according to the idle state of the peer node as stored in the Session Context of the session (either the [Secure Session Context](#) or the [Unsecured Session Context](#) depending on the Session Type). For all subsequent messages of the exchange, the base interval, *i*, SHOULD be set according to the active state of the peer node as stored in the Session Context of the session (either the [Secure Session Context](#) or the [Unsecured Session Context](#) depending on the Session Type), unless the sender has other means to determine whether the device is active or idle. The backoff base interval SHALL be set to a value at least 10% greater than the idle interval of the destination:

- If [PeerActiveMode](#) in the Session Context is true:
 - *i* = [SESSION_ACTIVE_INTERVAL](#) of the peer
- Else the peer is in idle mode:
 - *i* = [SESSION_IDLE_INTERVAL](#) of the peer
- *i* = [MRP_BACKOFF_MARGIN](#) * *i*

The [MRP_BACKOFF_THRESHOLD](#) parameter creates a two-phase scheme which begins with linear